

Question One [36 marks]: This is a multiple-choice question. Please copy your answer to the table provided at the end of page 4.

1. Given the URL

`https://www.example.com/products/electronics/laptops.php?filter=discounted&sort=price_desc#reviews`, which part of the URL represents the path?

- ☒ A) `/products/electronics/laptops.php` ✓
- ☐ B) `/products/electronics/laptops.php?filter=discounted&sort=price_desc`
- ☐ C) `https://www.example.com` ✗
- ☐ D) `reviews` ✗

2. Forms submitted using a GET request can do all of the following **except**:

- A) Transmit user-entered data to the server.
- B) Display the query string in the URL. ✓
- ☒ C) Upload files as attachments to the server. ✗
- D) Result in pages with URLs that can be bookmarked ✓

3. Which of the following HTTP status codes is typically used to indicate that the server could not find the requested resource?

- A) 500 Internal Server Error
- B) 403 Forbidden
- C) 301 Moved Permanently
- ☒ D) 404 Not Found

4. Why does the browser need to resolve the domain name into an IP address before making an HTTP request?

- A) To ensure the server is secure and trustworthy
- ☒ B) To know the exact location (IP address) of the webserver to send the HTTP request to
- C) To determine which content type (HTML, JSON, etc.) is supported by the server
- D) To optimize the page load time by caching the URL

5. In the following form, what is the size of the `$_POST` array when the form is submitted?

```
<form action="process.php" method="post">
  <input type="text" name="fname" size="10" />
  <input type="text" id="lname" size="10" />
  <input type="submit" />
</form>
```

- A) 2
- ☒ B) 1
- C) 3
- D) 4

6. Which HTTP header is typically used to specify the **expiration time** of a cached resource?

- A) Content-Length
- C) Expires

- B) Last-Modified**
- D) Content-Type

7. What is the output of the following script after the 2nd request, and what is the value set to the month cookie?

```
<?php
$m = ["Jan"=>5, "Feb"=>10, "Mar"=>10, 0=>"Feb"];
$m[] = "Mar";
if(count($_COOKIE) == 0) {
    setcookie("month", $m[1], time() + 3600);
} else {
    echo $m[$_COOKIE['month']] ;
    setcookie("month", $m[0], time() - 3600);
}
?>
```

- A) Output: 7, Cookie Value: Feb
- B) Output: 10, Cookie Value: The cookie is deleted.
- C) Output: 10, Cookie Value: Feb
- D) Output: Mar, Cookie Value: The cookie is deleted.

8. Which of these is a key advantage of using semantic elements in HTML?

- A) They enhance the webpage's visual appearance.
- B) They make the webpage compatible only with modern browsers.
- C) They reduce the need for CSS in styling the page.
- D) They improve accessibility and SEO by providing meaningful structure to content.**

9. You are given the following HTML code:

```
<form action="upload.php" method="post" enctype="multipart/form-data">
  <label>Upload Your CV</label>
  <input type="file" name="cv" />
  <input type="submit" />
</form>
```

In the upload.php script, you want to move the uploaded file to a directory named applicantCVs. Which of the following is the correct PHP code to achieve this?

- A) move_uploaded_file(\$_FILES['cv']['name'], 'applicantCVs/');
- B) move_uploaded_file(\$_FILES['cv']['tmp_name'], 'applicantCVs/' . \$_FILES['cv']['name']);**
- C) move_uploaded_file(\$_FILES['cv']['tmp_name'], 'applicantCVs/');
- D) move_uploaded_file(\$_FILES['cv']['name'], 'applicantCVs/' . \$_FILES['cv']['tmp_name']);

10. Which semantic element is best suited for displaying a captioned image?

- A) <figure> B) <main> C) <footer> D) <article>

11. Which of the following is the correct HTML code for a password input field with the following requirements: a minimum length of 6 characters, a maximum length of 12 characters, and the password must end with a digit?

- A) <input type="password" name="password" minlength="6" maxlength="12" pattern="[a-zA-Z0-9]*\d" required>
 B) <input type="password" name="password" minlength="6" maxlength="12" pattern="{6,12}\d" required>
 C) <input type="password" name="password" minlength="6" maxlength="12" pattern=".*\d\$" required>
 D) <input type="password" name="password" minlength="6" maxlength="12" required>

12. What is the output of the following PHP script if the user selects the "Second Option"?

```
<html> <body>
<?php
$marks_ = ['first'=>[90,85,80],
           'second'=>[94,90,85]];
if (isset($_POST['option']))
    foreach($marks[$_POST['option']] as $value)
        echo $value . " ";
?>
<form method = "post" action = "<?php echo $_SERVER['PHP_SELF']?>">
<label> Choose an Option </label> <br/>
<input type="radio" name = "option" value = "second"> First Option</input><br/>
<input type="radio" name = "option" value = "first"> Second Option</input><br/>
<input type= "submit"/>
</form>
</body> </html>
```

- A) 94 90 85 B) 80 85 80 C) 85 90 94 D) 90 94 85

Copy your final answers for the multiple-choice questions into the table provided. Ensure you write your answers using capital letters only.

1	2	3	4	5	6	7	8	9	10	11	12
A	C	D	B	B	A	A	D	B	A	A	A

8 x 3 = 24

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Question Two [20 mark]

Create an HTML5 static page for a car rental service. The web page should allow users to search for available rental cars based on specific criteria. The form must include the following input fields:

- Car Type: Use a dropdown menu to allow users to select the car type (e.g., Sedan, SUV, Truck), the Car Type field is required. *(select name or value)*
- Price: Include a number input field for the rental price. The Price field only accepts numbers within the \$50 to \$500 range. *input = number min 50 max 500*
- Car Features: Use checkboxes to allow users to select multiple car features (e.g., GPS, Sunroof, Heated Seats). *input = check*
- Rental Start Date: Include a date picker so the user can select the rental start date. *(date) input = date*
- Rental End Date: Include a date picker so the user can select the rental end date. *input = date*
- The form must include a Submit button to send the form data and a Reset button to clear the form. *button type = submit type = reset*
- Using the post method, the form should be submitted to the "search.php" script located in the "carRent" folder on the "ajar.com" domain.
- Add labels for all input elements.

<!DOCTYPE HTML>

<html lang="en">

<head>

<title> Car rental service </title>

</head>

<body>

<form method="post" action="https://ajar.com/carRent/search.php">

<fieldset>

<label> select car type </label>

<select name="Type" value="CarType">

<option name="option" value="Sedan">Sedan </option>

<option name="option" value="SUV">SUV </option>

<option name="option" value="Truck">Truck </option>

</select>

<option type="CarType">Sedan </option>

<option type="CarType">SUV </option>

<option type="CarType">Truck </option>

</select>

</fieldset>

</body>

</html>

</fieldset>

~~<input type="number">~~

<label> select range </label>

<input type="number" min="50" max="500" />

</fieldset>

<fieldset>

<label> select feature </label>

<input type="checkbox" name="check" value="Feature[]" > GPS </input>

<input type="checkbox" name="check" value="Feature[]" > sun roof </input>

</fieldset>

<fieldset>

<label> select start date </label>

<input type="date" />

</fieldset>

<fieldset>

<label> select end date </label>

<input type="date" />

</fieldset>

~~<input type="submit" />~~

<input type="submit" />

<input type="reset" />

</form>

</body>

</html>

Question Three [20 mark]

Develop a simple shopping cart system for an e-commerce website. The product data must be retrieved from a MySQL database and displayed dynamically in an HTML table. The products relation is given as follows; the relation key is underlined:

Products(id:integer, name:string, price:real, desc:string).

Assume the database is created on the localhost server with the name eStore, the username admin, and the password comp@334.

A) Write a PHP script config.inc.php to connect to the MySQL database using PDO. Use the provided credentials to establish the connection and include it in other PHP scripts you will write in Question Three and Question Four.

B) Write a PHP script (products.php) that dynamically creates an HTML page by fetching product data from the products database table and displaying it in an HTML table. The table should include the following columns: product ID, Product Name, Description, Price, and an "Add to Cart" link for each product should send the product ID to cart.php via GET.

config.inc.php

```
define ("Host", "localhost");
define ("db", "estore");
define ("uname", "admin");
define ("upass", "comp@334");

define ("ASS", "host=" . Host . "DB = " . db);

PDO ("ASS", "uname", "upass", "UPASS = " . upass);
```

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Question Four [24 mark]

Manage the shopping cart, continue from question three. Write a PHP script (cart.php) that uses PHP sessions to manage the shopping cart as follows:

- Add items to the cart when a product is selected. If an item is already in the cart, increment its quantity. The product is selected when the user clicks the "Add to Cart" link, as in Question Three.
- Display the cart in an HTML table, showing the Product Name, Quantity, Price, and Total Price, and a "Remove" link that removes an item from the cart.
- Do not re-write the script that connects to the database; include config.inc.php from Question Three.

```
<?php>
include "config.inc.php";
start_session();

if (method [post] - post) {
    sql = update on store where id = $id;
    $pdo->query(sql);
}
```

```
<table border="1">
<tr>
<th> product Name </th>
<th> quantity </th>
<th> price </th>
<th> total price </th>
</tr>
```

```
<tr>
<td> <?php $Get[$name]?> </td>
<td> <?php $Get[$quantity]?> </td>
<td> <?php $Get[$price]?> </td>
<td> <?php $Get[$total price]?> </td>
</tr>
</table>
```